

535514 Reinforcement Learning

Pre-Lecture Assignment: Lecture 6

Instructor: Ping-Chun Hsieh

Email: pinghsieh@nycu.edu.tw

Problem: GAE Implemented in Python

In Lecture 5, we just learned the GAE estimator, under a proper model-free prediction method that predicts the value function. In Lecture 6, we will further discuss how to get GAE under a value function approximator. Let's try to match the concept of GAE with the actual Python implementation. You can refer to the slides **pp. 3-14** of Lecture 6 as a starting point when answering the questions below.

- 1) GAE:** The `gae_nn.py` is a Python file generated by ChatGPT. Take a read through the code. Can you explain the purpose of Lines 25-29?
- 2) A deeper look at the code:** Can you convert each line of Lines 48-58 into math equations?
- 3) Do you fully agree with ChatGPT?:** Can you either identify 1 bug in the code or provide at least 1 suggestion for improving the code?

Submission Guidelines and Remarks

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.
- Please submit it by 9pm, 3/24 (Monday).
- If you have any questions, feel free to post them on Slack.

